

Date: 18th JUNE 2022

To,

BBELL INDUSTRY LLP.

Shaykha, Gujarat,

SUB: Technical cum Commercial Offer for Membrane Filter Press

1000 x 1000 mm MIX PACK

ENQ REF: Email/ 6 June 2022

OFFER RVS: BBEL/1000/MEM/170622/R0

KIND ATTN: Shri Kanak Bhai J. Pansari.

Managing Director.

Dear Sir/Madam

This is with reference to your valued inquiry regarding above subject and reference, we are pleased to submit our techno commercial & offer as per following MOC and details.

Annexure – 1	:	Design, Configuration and MOC.
Annexure – 2	:	Price.
Annexure – 3	:	Terms & Conditions.

Special Note: -

1. This offer is made in anticipation that working conditions are normal, ambient and non-hazardous.
2. Shown equipment in offer are for demo purpose only. Actual equipment may differ as per client requirements & design.
3. The quoted offer is exclusively meant for the recipient only and it is restricted to be circulated for what so ever reason without written consent.
4. This is our standard scope of supply.

We trust our offer is in line with your requirements and looking forward to the pleasure of receiving your valued order at the earliest for same.

KETAN KANSARA

PARTNER

ANNEXURE- 1 Design, Configuration & MOC



MEMBRANE MIX PACK FILTER PRESS

1. Model No. :	F1510-DCC-2840-M
2. No of Equipment require:	01 Nos.
3. Size & type of Filter Press:	• Size 1000 x 1000 mm – Semi Auto Filter Press • Membrane Mix pack Filtration Element. • CLOSE Configuration. • Center Feed. • Individual water Squeezing System. • Manual Cake Discharge.
4. Design/working Pressure :	17 / 15 bar
5. No. of Chamber :	28 chambers per press OPTIONAL- Distance Piece for 14 chambers for future expansion.
6. Number of plates in press:	• 14 Nos. – Membrane Filter Plate • 13 Nos. – Companion Filter plate • 1 No – First Plate • 1 No – End Plate • 29 Nos. – Total

7.	Cake Thickness :	40mm (Before Squeezing.)
8.	Batch Volume :	739.2 Liter /batch @ 26.4 Liter/Plate.
9.	Total Filtration Area :	43.68 Sq.mt/press @ 1.56 sq.mt/Plate
10.	Filter Press Structure: 	• Structure: Fabricated & Welded. • MOC: Heavy Duty Mild Steel • Make: AMNS (ESSARE) make/ equivalent. • Grade: IS-2062 (ASTM A36) • Design: 16 bar with safety factor of 1.2. • Tie Bar: MS Rectangle Solid flat as Tie bar with Corner Radius on Load Suffering Junction. • Welding: ASME Standard Fillet Weld with Flux Coated GMAW/MIG Welding. • Cladding: SS 304 x 2mm thick "U" cap at top side of the tie bar.
11.	Type of Locking : 	• Cylinder Design: Double Acting - Front Flange Mounted Non-Cushioned. • Capacity: 150 Ton/each • Stroke Length: 600mm • Ram: MOC-EN-8 duly Hard Chrome Platted. Protected by Leather Bellow. • No of Cylinder: 1 No. • Working pressure: 210 kg/cm² max. • Design Pressure: 280 Kg/cm² • Seals: U Cup Seal + Chevron Rod Seal + Wiper Seal. Ultra Filtech filter press are designed with low pressure hydraulic system i.e. cylinder are bigger in diameter and at low hydraulic pressure it gives the required tonnage.
12.	Hydraulic Power Pack : 	• Make: Customize • Design Pressure: 300 kg/cm² • Working Pressure: 250 kg/cm² • Tank Capacity: 150Ltr • Power: 2.2 kw • Pump Set: High pressure dual pump set. • Valves: POLYHEDRON/YUKEN/ equivalent makes Pressure Control Module + SOV operated Direction control valve + pressure release valve + pressure relief valve + Pressure switch + pressure gauge + Oil filter + oil level indicator and set of high pressure set of hose pipe. • Electric Motor will be BB/CG/Havells/equivalent makes + NFLP. • Operation: Full Automatic Solenoid Operated filter press OPEN/CLOSE operation with auto compensator hydraulic circuit. • Hydraulic Oil will be in client scope.

13. Filter Plates: 	• Size: 1000 x 1000mm • Model: OPTIMA • Type of Plate: Mix Pack Diaphragm Filter Elements. • Feed Position: Center Feed. • MOC of Plate: <u>Polypropylene</u> for Membrane Body Plate & Companion Plate. • MOC OF Diaphragm: TPE - Detachable - Membrane (diaphragm). • Filtration Surface: Smart Drain Pips. • No. of Stay Bosses: 4 nos. • Cake Thickness: 40mm $\pm$2 • Arrangement: Drilled to adopt Cross Cake washing & cross cake airing. • Max. Working pressure: 15 bar below 60°C • Max. Operating Temp.: 85°C below 4 bar.
14. Nozzles & Feed Head Manifold. 	• Feed Nozzle: Size-3"nb, MOC-PP, Flanged end, Qty-1 no. • Discharged Nozzles: Size-2"nb, MOC-PP, Flanged end, Qty – 4nos. • Feed Header: Size-3"nb, + Feed Pressure Gauge. • Discharged Header: 3"x 2"nb, MOC-PPR-PN10, Header with Cross Cake Washing and Cake Airing Facility. • Cake Airing Header: 1"nb, MOC-PPR, header with Valves. • Valves: a. Feed valve will be 3"nb PP Ball valve. b. Discharge valves will be 2"nd, PP ball valve. c. Wash valve will be 1.5"nb – PP ball valve. d. Air valves will be -1"nb – PP Ball valves e. All valves will be Hand operated manual valves. f. All flanged duly drilled to ASA#150.
15. Squeezing System: - Individual Water type.  	• Squeeze Pump: CRI/LUBI/WILO/Equivalent make High pressure multistage centrifugal pump. • Pump Qty.: - 1 No. (working) • Squeeze Header: SS304 header with individual hose connection + Pressure Gauge + Pressure Switch. • Working Pressure: 16 bar max. • All fittings will be threaded to BSPT Standard. Squeezing Skid: • Platform: MS. Epoxy Painted. • Tank: MOC- HDPE/PVC. • Pump: Squeeze pump will be installed on same skid. • Manifold: Tank Outlet to pump inlet, PVC piping with Manual valves + fittings + Pressure gauges + NRV + safety valves are included. Note: 1. All filed piping will be in client scope.

16. Filter Cloth:	• MOC-PP • Multi Filaments duly heat set and calendared type. • Having 5-8 micron and above retention capacity. • Qty. – 1 Set (installed)
17. Control Panel:	• Cabinet: BCH/ELDON make IP-55 rating electrical enclosure as electric cabinet. • Size: suitable • Switchgears: Schneider • Control voltage: 230 VAC. • Cable entry: Bottom
18. Feed Pump:	• Client Scope
19. Hydraulic Oil for Power Pack	Client Scope – SERVO/ENCLO - 68
20. Structure Paint :	• Surface Preparation: SSPC-SP-10. (SA 2 ½. In-house facility) • Primary Coat: One Coat of F9 @65-75 micron DFT/coat • Intermediate Coat: None • Finish Coat: One Coat of F2 @40 micron DFT/coat • Total DFT: 105-115 microns • Color shade: EPOXY -RAL-5002

ANNEXURE - 2 PRICE BID

As per Annexure 1 following will be the *prices for one filter press.*

NO.	EQUIPMENT DESCRIPTION	QTY.	TOTAL PRICE In INR
1.	MEMBRANE MIX PACK FILTER PRESS MODEL NO- F1510-DCC-2840-M + Size-1000x 1000 mm- <u>Semi Automatic Filter Press.</u> + Heavy duty Filter Press structure of 15 bar max working Pressure. + Installed with 28 chambers of Mix Pack type Filter Plates pack of 40mm Cake Thickness. + Hydraulic Closing System. + Set of Nozzles. + Feed Head Manifold with Manual valves. + External Water Squeezing System on Skid. + 1 set of Filter Cloth + Control Panel	1	25,10,000=00
2.	Provision of 14 Chambers for future expansion - Extra	1	52,500=00
3.	Packing & Forwarding + Transportation		Extra at Actual
4.	Supervision of Erection & Commissioning charges – Rs. 7500/day		Extra at Actual

Note: Above price are without any flameproof consideration if not specified.

ANNEXURE – 3

TERMS & CONDITIONS

1. PRICES

1. Prices are FOR our Ex-works ULTRA FILETCH, Navsari.
2. Packing cost 2% will be extra.
3. Taxes will be 18% IGST extra or as per ruling government rules.
4. Freight, Transit Insurance, Octroi etc. to be arranged by you.

2. PAYMENT

1. 30% advance of total order value along with clear PO.
2. 60% against PROFORMA Invoice before delivery of each equipment.
3. Balance 10% after erection and commissioning or 3 months from the date of supply against PBG of 18 months from the date of supply or 12 months from the date of commissioning whichever is earlier.
4. In the event of delay in lifting of equipment after 2 weeks of inspection or call of readiness of the equipment, Rs. 25,000=00 per week will be charged as storage charges.

3. DELIVERY

Our normal delivery is **10-12 weeks**, but exact delivery will be quoted at the time of finalization of order. Delivery period will begin after fulfillment of following conditions

1. The drawings are approved from your end.
2. All technical specification and requirement are finalized.
3. Commercial terms are clear and advance received.

4. ERECTION & COMMISSIONING.

Erection & commissioning of equipment will include:

1. Only supervision of the main equipment erection with filter plate loading. Fitment of all other assemblies other than main structure & filter plate loading will be done by supplier.
2. Training to the personnel will be given with all the necessary knowledge required for the operation of the filter press.
3. Any extra days other than informed in offer will be charged at 1.5 times the contract rate.
4. Lodging + ground conveyance to site during erection & commissioning will have to be provided by buyer.

5. INSPECTION

Final inspection of equipment will be given at our end with prior intimation. Following test will be given during inspection.

1. Visual inspection.
2. Filter Press opening/closing hydraulic pressure test.
3. Plate shifting test in case of Plate shifter ordered.
4. Cloth washing dry cycle in case of cloth washing ordered.
5. PLC Dry Cycle demonstration in case of automation.

6. WARRANTY

1. The period of warrantee is 18 months from the date of invoice or 12 months from the date of commissioning and stops when fulfilling the term, although the installation for whatever reason has not been done.
2. ULTRA FILTECH assures the good quality of the component materials used and installed. In case of any faulty part, ULTRA FILTECH will first try to repair and then will replace if required. In such event ULTRA FILTECH will not be liable for production loss. The part replaced in guarantee will be property of ULTRA FILTECH.

3. The guarantee does not include cases of the natural deterioration, the damages caused by the bad use or the negligence of buyer.
4. This warranty further excludes all parts subject to wear and tear and limit to shelf life of particular part.
5. The warrantee does not cover all the flaws caused by the transport.
6. **The warrantee does not include quality of filtration as final filtration depends on many parameters and the type of cloth used by buyer.**

7. ARBITRATION

All disputes and differences arising out of or connected with the order will be amicably settled by authorized representatives of both the parties. The jurisdiction will be Navsari (Gujarat).

8. VALIDITY OF THIS OFFER: Above offer is valid for **2 Weeks** from the date of offer.

9. ORDER: Order to be placed on ULTRA FILTECH.

10. Exclusions in Offer

1. Feed Pump & Cake Wash Pump.
2. First fill of hydraulic oil.
3. Works of electrical system beyond filter press.
4. The civil work and foundation required for machine.
5. Erection of basic structure and loading of filter plate into the press.
6. The Cranes for unloading at your end and positioning to your location.
7. Anything which is not mentioned in above offer.

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